What to Do When You Win the USAID DIV Fund: The Complete Technical, Financial, and Operational Roadmap (Start-to-Finish)

AgroBiometrix Implementation Protocol: A 24-Month Master Execution Plan from Award Negotiation to DIV Stage 2 ($1.5M) Scale-Up

***Award Mechanism**: USAID DIV Stage 1 Proof of Concept ($150,000 / 24 Months) • **Target Reach**: 10,000+ Smallholders*
***Lead Implementing Organization**: AgroBiometrix Engineering Unit / Naysa-84 Labs in consortium with FUNAAB, OOU, FPI, and UGA*
***Principal Investigator / Project Director**: Dr. Young Imienwain Irivboje (FUNAAB / Naysa-84 Labs)*

EXECUTIVE MANDATE: TURNING $150K SEED EVIDENCE INTO A $1.5M SCALING ENGINE

The Fundamental Philosophy of USAID DIV: Winning a USAID DIV grant is fundamentally different from receiving traditional donor aid or charity. DIV operates as an **evidence-driven venture fund** within the US Government. USAID is investing $150,000 into AgroBiometrix to answer one decisive question: *'Can an Africa-first, sovereign biometrical operating system with last-mile vernacular voice dissemination produce statistically verified yield and revenue gains among smallholders at a dramatically lower unit cost than traditional extension?'*
If you prove this with gold-standard causal evidence (Difference-in-Differences), rigorous unit economics ($0.02 radio / $0.20 voice robocall), and operational fidelity, AgroBiometrix unlocks the gateway to **USAID DIV Stage 2 ($1,500,000)** and nationwide government adoption. This master playbook defines your exact roadmap from Day 1 to Month 24.

Phase 0: Pre-Award Due Diligence & Negotiation (Months -2 to 0 / Post-Notification)

Upon receiving notification of selection from the DIV Agreement Officer (AO), the formal negotiation and pre-award compliance phase commences. This phase typically spans 60 to 90 days before initial funds are disbursed.

Key Action Items & Requirements:

- **SAM.gov & Unique Entity Identifier (UEI) Validation**: Ensure active registration on the US Government System for Award Management (SAM.gov). Naysa-84 / AgroBiometrix must maintain valid NCAGE and UEI codes.
- **Non-US Organization Pre-Award Survey (NUPAS) / Financial Risk Review**: USAID evaluates internal financial management, segregation of duties, procurement policies, and anti-fraud protocols.
- **Award Mechanism Selection (Fixed Amount Award vs. Cooperative Agreement)**: Stage 1 DIV awards are predominantly structured as **Fixed Amount Awards (FAAs)**. Under an FAA, disbursements are *milestone-based* rather than reimbursement-based. You are paid tranches upon verification of delivered milestones (e.g., approved MEL plan, baseline completed, vernacular audio engine deployed).
- **Designated Bank Account Establishment**: Open a segregated, dedicated institutional bank account (USD and local currency NGN) exclusively for USAID project funds. Never commingle grant funds with personal or general organizational capital.
- **Assignment of Key Personnel & Agreement Officer Representative (AOR)**: Finalize the formal designation of Dr. Young Irivboje as PI / Project Director and establish weekly communication protocols with your assigned USAID AOR in Washington.

Special Directive: Step-by-Step SAM.gov & NCAGE Guide for Nigerian Entities

Federal regulation (2 CFR Part 25) strictly mandates that foreign recipients of US grants hold an active registration in **SAM.gov** with a **Unique Entity Identifier (UEI)**. SAM.gov replaced the legacy DUNS number in April 2022. **Registration is 100% free of charge; beware of commercial third-party scams charging $500–$2,500.**

Registration Step & Field	Exact Input / Action for Naysa-84 Labs / AgroBiometrix	Strategic Rationale & Critical Notes
Step 1: NCAGE Code Request (NATO NSPA Portal) eportal.nspa.nato.int	• Enter Legal Name & Physical Address **identical to CAC documents**. • Global Location Number (GLN): **LEAVE BLANK**. • ISIC Code: 7210 (R&D; in Natural Sciences & Agriculture). • NACE Code: 7219 (Research & Experimental Development). • NAICS Code: 541715 (R&D; in Life Sciences) / 541511 (Software). • Common Procurement Vocabulary (CPV): **LEAVE BLANK**.	Non-US entities must obtain an NCAGE code before SAM.gov will validate their record. Codes are issued within 2–5 business days at **zero cost**. Entering GLN or CPV will trigger unnecessary document requests.
Future Business & SAM Question (NCAGE Questionnaires)	• *'NCAGE code needed for invitation to tender?'* → **Select NO**. • *'The NCAGE code is requested by SAM?'* → **Select YES**.	Selecting 'Yes' to tender forces you to provide military bid numbers. Selecting 'Yes' to SAM triggers an automated data bridge syncing NATO with the US Defense Logistics Agency (DLA) and SAM.gov.

Step 2: Login.gov Account (Administrative Credentials)	• Register with dedicated entity email: n84labs@gmail.com. • Activate Multi-Factor Authentication (Google Authenticator / SMS). • **Record and store the 16-character Emergency Recovery Key.**	Entity-dedicated Gmail accounts are fully accepted and prevent administrative lockout when personal or student university email accounts expire. The 16-character key is required if your device is lost.
Step 3: Document Readiness & Exact-Match Rule	• CAC Certificate of Incorporation / Business Name Registration. • CAC Status Report / Form CAC 1.1 showing official registered address. • Official Nigerian Bank Statement showing identical company address. • Federal Tax Identification Number (TIN) from FIRS.	**The Exact-Match Rule:** SAM.gov compares inputs against government registries. Discrepancies in spelling, punctuation, or suite numbers trigger manual reviews that cause 3–4 week delays.
Step 4: SAM.gov Validation & UEI Generation	• Select: **'Financial Assistance Awards Only'**. • Enter verified legal name, NCAGE, and physical address. • System automatically assigns your **12-character alphanumeric UEI**.	Financial Assistance registration is streamlined for grants and cooperative agreements, avoiding the complex FAR clauses required for US military commercial defense contracts.
Step 5: GSA Notarized Letter & Annual Renewal	• Prepare GSA 'Template for Non-US Entities' on organization letterhead. • Appoint Dr. Young Irivboje as authorized Entity Administrator. • Stamp and sign before a Nigerian Notary Public / High Court Commissioner. • Upload PDF to Federal Service Desk (FSD.gov) for active status.	Mandatory anti-fraud control. Once approved, status becomes **ACTIVE**. Valid for exactly 365 days; must be renewed annually via a 10-minute online click-through to remain eligible for disbursements.

Phase 1 to 5: The 24-Month Chronological Execution Lifecycle

Project execution spans five structured phases across two full agricultural cropping cycles (Wet and Dry seasons):

Timeline	Phase & Focus	Core Technical & Operational Deliverables	Disbursement Trigger
Months 1–4 Phase 1	**Inception, Governance & Baseline Survey**	• AOR Kickoff Conference & Approval of Year 1 Work Plan. • Finalize Monitoring, Evaluation & Learning (MEL) Plan. • Secure Institutional Review Board (IRB) ethical clearance (FUNAAB). • Register Pre-Analysis Plan (PAP) on AEA / OSF Registry. • Deploy digital enumerators (ODK/Kobo); complete Baseline Survey of N=3,000 smallholders across 30 agrarian clusters.	**Milestone 1** Approved MEL Plan + IRB Clearance + Baseline Dataset ($35,000)
Months 5–8 Phase 2	**Technical Engineering & Broadcast Setup**	• Production deployment of Suite 8: Last-Mile Vernacular Radio Engine. • Integrate Africa's Talking API for 2G GSM cellular voice robocalls. • Onboard 10–15 rural community FM radio stations (ZaraRadio/Rivendell automated RSS podcast polling). • Ingest NARS soil acidity titration & SAMMAZ maize trial datasets. • Complete colloquial translation & voice QA (Hausa, Yoruba, Swahili, French).	**Milestone 2** Verified Telephony & Radio Pipeline + Station MoUs ($30,000)
Months 9–14 Phase 3A	**Season 1 Rollout (Major Wet Season)**	• Launch dawn/dusk FM radio advisories (lime neutralization + SAMMAZ-15 seed). • Dispatch targeted 2G GSM voice robocalls to 1,500 treatment smallholders. • Real-time telemetry monitoring (call completions, radio transmission logs). • Mid-season crop health monitoring & Fall Armyworm alert broadcast. • Quarterly Progress Report (QPR 1 & 2) submitted to USAID AOR.	**Milestone 3** Season 1 Broadcast Logs + 10k Farmer Reach Audit ($30,000)
Months 15–18 Phase 3B	**Season 2 Rollout (Minor Wet / Dry Season)**	• Deploy dry-season irrigation advisory via FAO-56 Penman-Monteith (ET0). • Broadcast hermetic storage post-harvest alerts & livestock THI heat stress. • Conduct intermediate pulse surveys to measure input purchase and lime adoption. • Annual Review Conference with USAID DIV team; submit Annual Report.	**Milestone 4** Annual Performance Report + Season 2 Deployment ($25,000)

Months 19–21 Phase 4	**Endline Survey & DiD Econometric Analysis**	• Execute Endline Survey of all 3,000 farming households (target attrition < 5%). • Estimate causal treatment effect (δ) via Difference-in-Differences (DiD). • Calculate verified Benefit-Cost Ratio (BCR) and cost-per-farmer-contact. • Draft publication-grade evaluation manuscript for academic journal submission.	**Milestone 5** Econometric Evaluation Report + Clean Datasets ($20,000)
Months 22–24 Phase 5	**Closeout, Dissemination & Stage 2 Preparation**	• Host National Policy Dissemination Workshop in Abuja with FMAFS and NARS. • Upload de-identified datasets to USAID Development Data Library (DDL). • Publish technical evaluation on USAID Development Experience Clearinghouse (DEC). • Final Performance Report (FPR) and financial audit sign-off. • **Submit USAID DIV Stage 2 Application ($1,500,000 Scaling Grant)**.	**Milestone 6** Final Closeout Report + DEC/DDL Submission ($10,000)

Detailed $150,000 Budget Allocation Across the 24-Month Grant

Every dollar of the $150,000 grant must be strategically allocated across technical engineering, field evaluation rigor, last-mile voice delivery, and compliance governance:

Budget Category & Line Item	Allocation ($)	% Total	Strategic Operational Justification & Description
1. Personnel & Scientific Leadership • Principal Investigator / PI (Dr. Young Irivboje) - 25% LOE • Co-Investigators (Agronomy, Economics, Breeding) • Full-Stack Cloud/Telephony AI Software Engineer (pt)	**$42,000**	**28.0%**	Covers key personnel Level of Effort (LOE) to direct operations, supervise multi-agent engineering, ensure biometrical rigor, oversee field enumerators, and lead econometric modeling across the 24-month grant period.
2. Field Surveys & Impact Evaluation • Baseline Survey (N=3,000 households, 30 clusters) • Endline Survey (N=3,000 households with tracking) • 20 Agricultural Enumerator Stipends & Training • Field Supervisors, Tablets, Mobile Data & ODK setup	**$45,000**	**30.0%**	The core evidence engine of Stage 1. Covers enumerator training, field logistics, household surveys, GPS mapping of trial clusters, farmer respondent time compensations, and data quality audits.
3. Telephony, Voice & Vernacular Audio Engine • Africa's Talking Outbound 2G GSM Voice Airtime • 10–15 Community FM Radio Partnership Broadcast MoUs • Vernacular Voice Recording, Dialect QA & Studio Fees • Cloud Storage & Automated RFC 2.0 RSS Hosting	**$28,000**	**18.7%**	Funds ~60,000–80,000 outbound cellular voice robocall minutes to smallholder basic feature phones, dawn/dusk broadcast slot partnerships with 10–15 rural FM stations, and vernacular voice talent.
4. Field Logistics, Travel & Workshops • Agrarian cluster community town halls & farmer meetings • Inter-state travel (Oyo, Osun, Ogun, Kaduna trial zones) • Vehicle fueling, rural mobility, field per diems • Extension agent / cooperative leadership briefing days	**$18,000**	**12.0%**	Enables physical monitoring, cluster sensitization meetings with cooperative heads, ground-truthing of soil acidity and Fall Armyworm outbreaks, and safe researcher field mobility.

5. Compliance, Ethics, Auditing & Closeout • Institutional Review Board (IRB) ethical clearance • Independent external financial audits (Year 1 & 2) • National Policy Dissemination Convening (Abuja) • SAM.gov, legal compliance, banking & administration	**$17,000**	**11.3%**	Guarantees bulletproof USAID audit readiness, independent accounting verification, ethical human research compliance, and high-visibility policy dissemination in Abuja with federal ministry leaders.
TOTAL GRANT OPERATING BUDGET	**$150,000**	**100.0%**	**Fully balanced budget satisfying USAID DIV cost-efficiency requirements.**

USAID Compliance, Financial Regulations & Reporting Requirements

Managing US Government funding demands strict adherence to federal regulations (2 CFR 200 and USAID ADS 303). Failure to comply can result in clawbacks, suspension, or debarment.

1. Mandatory Reporting Cycles:

- **Quarterly Progress Reports (QPRs)**: Submitted within 30 days of each fiscal quarter end. Must detail milestone progress, smallholders reached, audio syndication metrics, qualitative challenges, and corrective actions.
- **Quarterly Financial Reports (SF-425)**: Standard federal financial disclosure documenting expenditures against approved budget line items and tracking milestone completion tranches.
- **Annual Performance Reports**: Synthesizes annual agronomic trial progress, preliminary survey insights, and operational refinements.
- **Final Evaluation Report & Data Upload**: Submitted within 90 days of project closeout, accompanied by de-identified raw datasets deposited into the **USAID Development Data Library (DDL)** and public reports published to the **Development Experience Clearinghouse (DEC)**.

2. Strict Financial & Procurement Rules (Allowable vs. Unallowable):

- **Segregated Banking & Timesheets**: All grant transactions must run through the dedicated project account. All personnel charging time must maintain signed, auditable bi-weekly timesheets documenting actual Level of Effort (LOE).
- **Unallowable Costs**: US federal regulations strictly prohibit using grant funds for: luxury goods, alcoholic beverages, entertainment, lobbying government officials, purchasing real estate or unapproved motor vehicles, or paying customs duties/tariffs.
- **Competitive Procurement & Anti-Terrorism Vetting**: Any single service contract or equipment procurement exceeding $10,000 must follow transparent three-quote competitive bidding and undergo mandatory SAM.gov / OFAC sanctions screening.

The Grand Strategy: Transitioning Stage 1 Evidence into a $1,500,000 DIV Stage 2 Award

The ultimate purpose of DIV Stage 1 is not simply to conclude a 2-year project, but to position AgroBiometrix for **DIV Stage 2: Testing and Positioning for Scale ($1,500,000)** or institutional scaling with sovereign ministries. DIV explicitly favors Stage 1 grantees who prove their case.

THE STAGE 2 SCALING BLUEPRINT ($1.5M USD)

What USAID DIV Looks for in Stage 2 Applicants:

1. **Rigorous Causal Evidence:** An econometrically sound Difference-in-Differences evaluation proving that smallholders receiving AgroBiometrix advice increased net farm yields (+8–15%) and certified seed/lime adoption (+20 percentage points) relative to control clusters.
2. **Cost-Effectiveness Superiority:** Empirical proof that AgroBiometrix delivers advisory at ~$0.02 (FM Radio) and ~$0.20 (2G GSM Robocalls), outperforming physical extension ($15–$25 per contact) by orders of magnitude.
3. **Demonstrated Demand Signals:** 10,000+ farmers reached, high broadcast listenership, cooperative willingness-to-pay or government co-financing commitments.

With this evidence in hand by Month 22, you submit the $1.5M Stage 2 application to scale AgroBiometrix across 1,000,000+ farmers across Nigeria, Ghana, and Kenya.

Immediate Practical Checklist for the Principal Investigator

To ensure operational readiness when the award notification arrives, execute the following steps:

1. **Verify SAM.gov / UEI Status**: Confirm that your organization's registration on SAM.gov is active and compliant for federal awards.
2. **Formalize Consortium Sub-Agreements**: Draft preliminary Memoranda of Understanding (MoUs) with FUNAAB, OOU, FPI, and UGA clarifying exact roles and LOE.
3. **Draft the IRB Protocol Early**: Prepare the household survey informed consent protocols for expedited FUNAAB ethics review.

4. **Engage Telecom & Radio Partners**: Secure expressions of interest from Africa's Talking and 10 regional community radio stations.
5. **Maintain Technical Documentation**: Preserve the exact mathematical benchmarks, journal manuscripts, and whitepapers we have established.